V. Kumar, Microsoft

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Summary

Hand-on AI Architect with over 20 years of IT experience in transforming organizations through Generative AI, data engineering and advanced analytics. Demonstrated expertise in designing and implementing Gen AI solutions and modern data platforms,

Extensive global experience at Switzerland, Belgium, Mexico, Middle East, India, Canada and USA leading cross-functional teams and leveraging cutting-edge technologies (Azure Open AI), including cloud platforms (Azure, GCP, AWS), large language models (LLMs), Big data to deliver actionable insights aligned with business objectives.

Core Competencies

- Led the architecture, design, and implementation of scalable and innovative AI solutions, RAG based agentic AI workflows(AutoGen, LangGraph) across multiple high-impact projects and proofs of concept, leveraging Azure Data & AI services, including Azure OpenAI Service, Azure Document Intelligence, Azure AI Search, Large Language Models (GPT-4o and GPT-4), Azure App Services, Durable Functions, Azure SQL Database, and Application Insights.
- Enterprise Data Strategy Development and go-to market strategy.
- Modern AI & Data Stack Implementation (Azure Open AI, LLMs, Databricks, Spark, Python)
- Cloud Architecture (Azure, AWS, GCP)
- Building & Mentoring High-Performing Teams with latest Gen AI Skills

Qualifications

- MBA in Operations Management with Operations Research and MIS from IGNOU, Delhi, India.
- Bachelor of Technology from Nagarjuna University, Guntur, India.

Certification:

- Microsoft Azure Architect Technologies https://youtu.be/iEZ-uIGb8j8
- Microsoft Certified Azure AI Engineer
- Microsoft Certified Azure Data Engineer
- Published Author of best-selling book on Data & AI: https://www.amazon.com/Data-Labeling-Machine-Learning-Python/dp/1804610542
- TOGAF 9.1 Certified Architect
- Oracle Certified Master, Java EE5 Enterprise Architect (OCMJEA 5)
- Spark Certified developer from IBM
- Scrum Alliance Certified Scrum Master (CSM)
- AWS Certified Technical Professional

Skills:

Language	**Python 3.13**, PySpark, Java 1.8, Scala 2.10, C, PL/SQL, Shell script
Data & AI and Cloud	**Azure Open AI, LLMs, Azure AI search, Azure Doc Intelligence, Azure Blob Storage, Azure Databricks, MS Fabric, Azure Cosmos DB, Azure SQL, EventHub, Azure data Factory, Azure Data explorer, Apache airflow**, GCP Storage, GCP Big Query, Dataproc, Cloud Spanner, Hadoop, Apache Spark, HDFS, Hive, Pig, Kafka, Oozie, Zookeeper, Elastic Search 5.2, Apache SOLR, AWS S3, EC2, Lambda, One Ops, Kubernetes
Machine Learning	Linear Regression, Logistic Regression, Decision tree, Random Forest Support Vector Machines, Bayesian model, Clustering, Neural network Tensor flow, Azure Machine learning. Natural Language Processing

Java	JDK JEE, EJB, JSP, JSF, Servlets, Struts, Spring MVC Spring 2.5, Spring Boot, Micro Services, Spring REST, Spring DATA JPA, Hibernate 3.2, Sonar, XML, XSD, XSLT, RESTful, JSON.
Databases	Oracle, DB2, Informix, SQL Server, NoSQL (Cosmos, Cassandra, Mongo)
Development Tools	VS Code, IntelliJ IDEA, Eclipse, Jupyter Notebook, PyCharm, IBM Web Sphere, Oracle Web Logic, Tomcat RAD, TOAD, DBeaver, JBOSS Fuse
Source Control	Git, Subversion, Rational Clear Case, Code Collaborator, VSS
Platforms	Windows, Linux, AIX, UNIX, Solaris, MS-DOS, OS/400

Professional Experience:

Microsoft **01/2022- Present**

Principal Cloud Architect (Data & AI)/Director – Microsoft

AI Tool for Financial statements Review:

- Designed the RAG based workflow solution architecture for an AI-driven application to analyze and extract insights from complex financial data while automating the verification of tax and audit documents. (pdf and excel formats)
- Leveraged cutting-edge Azure technologies, including Azure OpenAI, Azure AI Search, and large language models (LLMs), to create a seamless user experience and reduce manual effort significantly.
- Delivered a transformative AI solution that enhanced accuracy and efficiency for finance and audit teams.
- Guided a team of AI engineers to develop the quality review system to automate the business rules validation for compliance and accuracy. The solution processes unstructured data, converting into structured meta data using Azure document Intelligence. It leverages Azure Open AI, Prompt engineering, LLMs- GPT-4o and GPT 4 for auditing of financial reports.
- Designed Metadata and prompts and stored in Azure SQL database for efficient retrieval and management. The system uses hybrid search techniques, chunking methods and parallel processing to optimize performance.
- Enabled Monitoring and Observability through Azure application insights.

Sustainability report with Generative AI:

- Developed an AI-powered sustainability reporting solution leveraging Open AI APIs, LangChain and Azure Durable functions to automate the generation and evaluation of sustainability reporting.
- Developed Agentic AI workflow for prompt engineering, parallel processing and state management ensuring high quality and compliance reports
- Integrated Ragas for evaluating narratives on RAG metrics like "response relevancy", faithfulness and factual correctness
- Utilized Azure AI search (hybrid search) for semantic document retrieval and azure blob storage for managing large datasets
- Delivered a scalable, efficient and future-ready architecture enabling cost and time savings for sustainability reporting
- Technologies Azure Open AI APIs, LangChain, Ragas, Azure datable Functions, Azure AI search and Azure blob storage, Python Fast API, pandas and Python docx.

Streaming Data Platform for General Motors:

- Led the end-to-end design and development of a large-scale, real-time streaming data platform ingesting millions of events from connected vehicles.
- Architected a robust solution to process and analyze vehicle telemetry data, enabling real-time notifications to drivers, contributing to operational efficiency and customer engagement.

- The platform drove $10M in revenue for General Motors, showcasing the impact of actionable insights derived from data.

Cognizant Technology Solutions 06/2011- 01/2022

Associate Director-Engineering

I am responsible for delivery and solution architecture, design of multiple projects, proof of concepts in Data & AI practice using python, Scala. Spark. Java, Big data, Kafka, AI/ML, Angular, Kubernetes and managing a team of senior managers and managers.

Client: Walmart, Bentonville, AR **12/2020– 01/2022**

Project: Enterprise Inventory **Principal Architect-Cloud, Big data, AI**

The Objective the project is to migrate the Enterprise Inventory data from Hadoop to GCP Cloud.

- Defined and executed Data Strategy with complex data eco-systems.
- Lead a team of experienced Data engineers for end-to-end delivery of the project.
- Hands on expertise in designing and building end-to-end data pipelines using GCP and Azure.
- Designed and implemented Spark streaming jobs to ingest the Enterprise inventory data to GCS and GCP Big Query from Kafka using Cloud Dataproc, Spark and Scala.
- Resolved complex issues in Spark streaming AVRO schema deserialization.
- Migrated Prod 9 Hadoop data to GCS and GCP big query using Batch shell scripts.
- Optimized Spark jobs performance by tuning the configurations and Dataproc cluster.

Environment: Cloud Dataproc, GCS, GCP Big Query, Spark, Scala, Kafka, Spark Streaming, Hadoop, HDFS, Hive, Shell script, CI/CD, Java, Maven, Looper, Concord

Client: Sam's Club, Bentonville, AR **08/2019 – 11/2020**

Project: Supply Chain Data Visibility **Principal Engineer-Cloud, Big data, AI**

The Objective the project is build supply chain Data Lake on Azure for generating near real-time reports and sales forecasting for better visibility to the business.

- Defined, architected, and delivered Data, Analytics and AI strategy and solutions.
- Hands on expertise in designing and building end-to-end data pipelines using Azure and GCP.
- Provided Solution architecture for Supply chain Data modernization on the Cloud.
- Designed and implemented Supply chain Data pipeline for ingestion of data from on-premises Informix, SQL Server, Teradata and DB2 to **Azure Data bricks Delta Lake** and **Azure SQL**
- Designed and implemented Data ingestion pipeline to ingest data to GCP Bucket and Big query using **Dataproc, Spark streaming, GCS, Big Query, Cloud Spanner and Cloud SQL**.
- Implemented **PySpark Jobs** and ADF pipelines for Data ingestion for 5000 tables.
- Implemented **Sqoop** jobs to extract data from **HDFS** and ingest to **AZURE Blob storage.**
- Implemented Workflow Orchestration using **Apache Airflow** for scheduling thousands of jobs.
- Architected Time series forecast model for forecasting the sales using FB Prophet model.

Environment: Azure Databricks, GCP Dataproc, Big Query, GCP Spanner, Cloud SQL, Cloud Spanner, Azure Data Factory, Blob Storage, Azure SQL, Cosmos DB, Apache Air flow, Python, PySpark.

Client: AT&T, Toronto, ON, CA **05/2018 – 07/2019**

Project: Enterprise Data Management Tools **Data Scientist**

The objective of the project is to build an intelligent customer-matching engine that can process millions of enterprise customer accounts from various legacy source applications.

ROLES AND RESPONSIBILITIES

- Extracted the 40 million Enterprise Customer accounts from 25 legacy sources into Oracle.
- Built the customer data pipeline using **Kafka** 1.0 with **60 partitions and 60 listeners to Oracle.**
- Normalized and cleansed the missing data using standardization rules with SQL and Java.
- Detected the blacklisted customer accounts using ML Classification models.
- Designed and developed the various name matching and scoring techniques using the **Levenstein distance, Edit distance, Jaro Winkler string matching algorithms.**
- Grouped the enterprise customer data using Apache **SOLR N-Gram Tokenizers and Filters**.
- Designed NLP solution to detect complex relationships between companies using **NLTK**.
- Created a Data curation lab to perform Data science experiments for matching.
- Containerized the Python microservice with Flask, Docker and deployed to Kubernetes.

Environment: Kafka Stream1.0, Apache **SOLR**, Zookeeper, AWS Cloud, **RESTful Microservices**, Java, JDK 1.8, **Spring boot**, Micro Services, Oracle, PL/SQL, Unix, Docker, Kubernetes, ML, Python.

Client: Wal-Mart, Bentonville, AR **05/2017 – 04/2018**

Project: Global Supplier Management **Big Data Architect**

The objective of the Global Supplier Management System is to migrate the Supplier and Contracts data from legacy supplier system (DB2) to Nextgen supplier system (Cassandra) using Camel and Kafka.

As part of this, Audit and Reconciliation tool is developed to find the data integrity and synchronization issues between the source (DB2) and target (Cassandra) systems.

ROLES AND RESPONSIBILITIES:

- Migrated legacy Suppliers system to NO-SQL DB Cassandra using Camel and Kafka.
- Designed and developed Audit tool to validate data consistency between systems.
- Ingested Legacy Suppliers Data from DB2 and Cassandra DB into HDFS using Sqoop and Spark.
- Automated the Sqoop and Spark jobs using Ooze workflow Scheduler.

Environment: Horton works Hadoop, HDFS, Sqoop, Hive, Spark, Scala, Cassandra, DB2, IBM MQ, Camel, AZURE Cloud, Java 1.8, SOLR, RESTful services, Python, NumPy, Scikit Learn.

Client: Wal-Mart, Bentonville, AR **08/016 – 04/2017**
Project: Enterprise Inventory **Big Data Architect**
designed Inventory processing framework to provide a central repository of all Walmart inventory visible in near real-time to consumers across various domains using Kafka, Spark streaming and Cassandra.

Client: Wal-Mart, Bentonville, AR **09/2014 – 07/2016**
Projects: HRO-Associate Master and Next CCI **Engineering Manager**
Design and delivery of a Global Associate Master using MQ and RESTful Micro services.

Client: JB HUNT, Lowell, AR **02/2013 – 08/ 2014**
Project: Order Management **Technical lead**
Responsible for the design to migrate /rewrite the Main Frame system into Java platform for Order capture and processing in JB Hunt using Apache Camel, JBoss Drools and web services, Elastic search and migrated data base from on premises SQL Server to Azure Cosmos DB.

Client: WAL-MART, Bentonville, AR **12/2011 – 01/2014**
Project: Anti-corruption Documentum **Technical lead**
As an EMC Documentum consultant, responsible to create a repository for Anti-corruption compliance.

Project: Global Online Fulfilment Solution **Technical Architect**
Technical Lead for Dashboard, Pack and Ship tracks of 15-member team and handled design and development of global online fulfilment application that would allow Wal-Mart store associate to perform picking and staging of items ordered through Walmart.com using Jersey RESTful web services.

Project: Store Number Expansion **Technical lead**
Technical lead to identify the impacts of store number expansion and implementation of the changes across thousands of Wal-Mart store application built in Java in an innovative approach.

Project: Web M Tracks **Technical Architect**
Technical Architect responsible for the redesign of the Presentation layer and developed a web-based System for SAM's Club Membership Registration using JSF.

WIPRO TECHNOLOGIES 06/2010-05/2011

Client: Best Buy, Minneapolis, MN **06/2010– 05/2011**
Project: Carrier Activations Platform **Technical Consultant**
Primarily responsible for the architecture and design of the middleware platform

TATA CONSULTANCY SERVICES 12/2000-05/2010

Client: Bank of America, Jersey City, NJ **07/2009 – 05/2010**
Project: Product Master Environment (PME)-Fusion **J2EE Architect**
Designed and developed a pipeline to process the reference raw data in Investment banking.

Client: Harvard Business School, Boston, MA **02/2008 – 06/2009**
Project: Faculty Information Platform Project (FIPP) **Technical Architect**
Led the architecture and design of common core components, at Harvard Business School.

Client: India National Centre of Ocean Information Services (INCOIS) **11/2006 – 01/ 2008**
Project: Tsunami early warning system **Technical Manager**
Led the 25 -person technical team to establish a Tsunami alert system for the government of India using predictive data modelling, Data stage, Java-based decision engine and SMS alerts.

Client: LMR-IT, Kingdom of Bahrain **12/2005– 10/ 2006**
Project: Expat Management System **Technical Lead**

Client: VeriSign, Broomfield, CO **05/2005 –11/2005**
Project: Prepay Customer Care GUI **Technical lead**
Led the Design of Customer care GUI for the **US Cellular phone Prepay system**

Client: TELMEX, Tijuana, Mexico **06/2004 – 04/2005**
CMM Level Software Process Automation **CMM Consultant**

Performed CMM level 3 assessment for TELMEX, a major **telecom** Organization **Client: Belgacom, Brussels, Belgium** **03/ 2003–05/2004**

Java Framework **Module lead**

Developed the Java Framework to develop internal billing applications within Belgacom (Proximus).

Client: Indian National Centre of Ocean Information Services (INCOIS) **03/2002–02/2003**

Ocean Portal (www.incois.gov.in**)** **Java lead**

Designed and developed a web portal (www.incois.gov.in) Govt. of India

Client: Credit Suisse, Zurich, Switzerland **12/2000–02/2002**

New Inter-company Booking Structure (NIBS) **Module lead**

Designed and developed a Securities clearing system for a Credit Swiss Private Bank (CSPB).